



Addis Ababa University

School of Earth Sciences

Remote Sensing and Geo-Informatics Stream

Mini Project

Geostatistical analysis and modeling

of

Ground Water Quality parameters

Borena zone, south Ethiopia

Course No: ERSC 694

Course Name: Spatial Analysis and Mapping

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1. Introduction

A clean drinking water is vital for a healthy life. Most of the earth's liquid fresh water is stored or found underground aquifer and groundwater is the major sources of drinking water in the Groundwater is the major source of drinking water. Most of the Earth's liquid fresh water is found, not in lakes or rivers, but is stored underground in the aquifers. Indeed, these aquifers provide a valuable base flow supplying water to rivers during periods of no rainfall. They are therefore an essential resource that requires protection so that groundwater can continue to sustain the human race and the various ecosystems that depend on it. The contribution from groundwater is vital; according to Morris and et.al, two billion people depend directly upon aquifers for drinking water, and 40 percent of the world's food is produced by irrigated agriculture that relies largely on groundwater. In the future, aquifer development will continue to be fundamental to economic development and reliable water supplies will be needed for domestic and irrigation purposes. Water stored in the ground beneath our feet is invisible and so its depletion or degradation due to contamination can proceed unnoticed, unlike our rivers, lakes and reservoirs, where drying up or pollution rapidly becomes obvious and is reported (Morris et al. 2003)

1.1 Water safety and human health

The quality of drinking-water is a powerful environmental determinant of health. Drinking-water quality management has been a key pillar of primary prevention for over one-and-a-half centuries and it continues to be the foundation for the prevention and control of waterborne diseases. Water is essential for life, but it can and does transmit disease in countries in all continents – from the poorest to the wealthiest. The most predominant waterborne disease, diarrhea, has an estimated annual incidence of 4.6 billion episodes and causes 2.2 million deaths every year. There are several variants of the fecal-oral pathway of water-borne disease transmission. These include contamination of drinking-water catchments (e.g. by human or animal faces), water within the distribution system (e.g. through leaky pipes or obsolete infrastructure) or of stored household water as a result of unhygienic handling. Millions of people are exposed to unsafe levels of chemical contaminants in their drinking-water. This may be linked to a lack of proper management of urban and industrial wastewater or agricultural run-off water – potentially giving rise to long term exposure to pollutants, which can have a range of serious health implications. Or it may be linked to naturally-occurring arsenic and fluoride, which cause cancer and tooth/skeletal damage, respectively. The Millennium Development Goal target 7c calls for reducing by half the proportion of people without sustainable access to safe drinking-water and basic sanitation by 2015. Reaching this target implies, inter alia, tackling both the quantity (access, scarcity) and quality (safety) dimensions of drinking-water provision.

1.2 WHO and the Guidelines for Drinking-water Quality

The World Health Organization (WHO) was set up in 1948 with the objective of promoting 'the attainment by all peoples of the highest possible level of health'. WHO has a wide range of functions, which include promoting (in cooperation with other specialized agencies) the improvement of nutrition, housing, sanitation, recreation, economic or working conditions with a bearing on health and other aspects of environmental hygiene. One of the main roles of WHO is to establish international norms to protect human health. Since 1958, as part of its activities on drinking-water and health, the Organization has published – at around ten-year intervals – several editions of International Standards for Drinking-water and subsequently, the Guidelines for Drinking water Quality.

The Guidelines for Drinking-water Quality is one of the longest-standing normative publications of WHO. They provide an evidence-based point of departure for standard setting and regulation as a basis for health protection. They include an assessment of the health risks presented by the various microbial, chemical, radiological and physical constituents that may be present in drinking-water. Where applicable, they derive maximum concentration guideline values for these hazardous constituents (table 1).

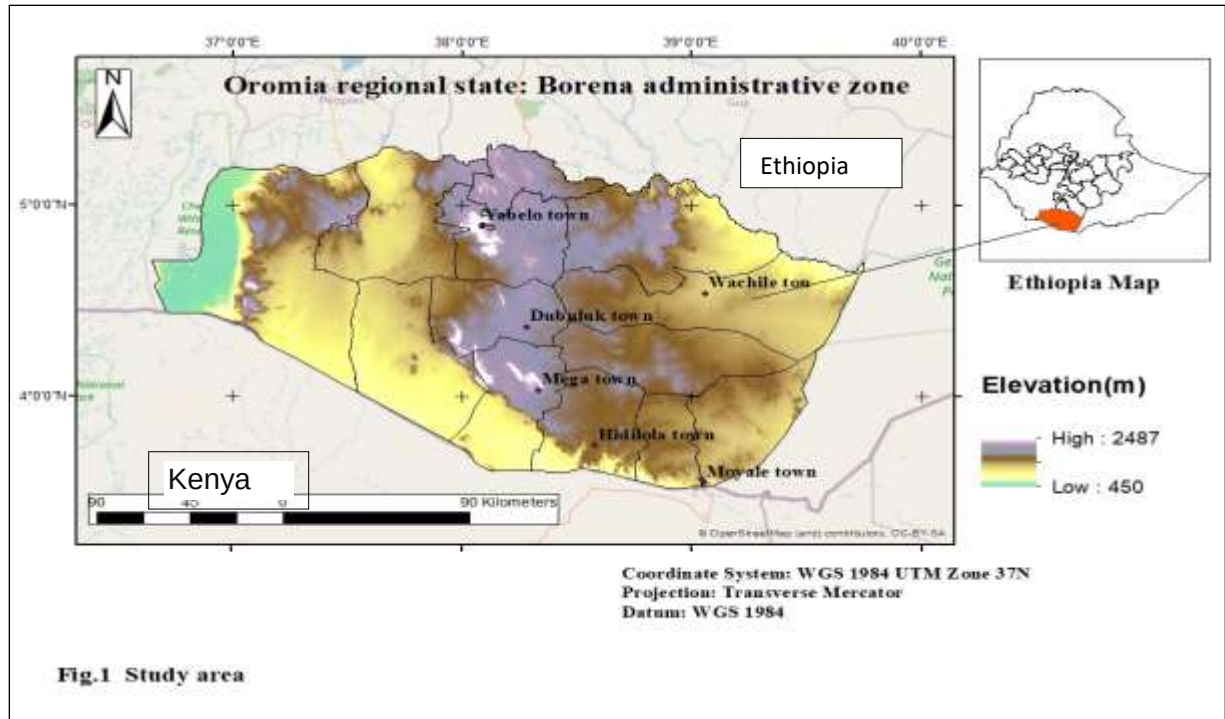
Table 1: Water quality standard(selected)

Parameter	Ethiopian Standard (2011)	WHO standard	Remark
pH	6.5 – 8.5	8.2 -8.8	4
EC (µS/cm)	1500	1,500 µS/cm	4
TDS (mg/L)	1000	500	5
Calcium (mg/L)	75		2
Magnesium (mg/L)	50		2
Sodium (mg/L)	200	50	3
Potassium (mg/L)	1.5		2
Chloride (mg/L)			
250 5			
Sulphate (mg/L)	250	250	5
Nitrates (mg/L)	50	50	5
Bicarbonate (mg/L)			1

2. Study area, Data and Methodology

2.1 Study area

Borana is one of the twenty three administrative zones in Oromia region and located between 3° 31' 41" and 5° 09' 31" latitude and 37° 01' 13" and 39° 32' 59" longitude at a distance of 570 kms south of Addis Ababa on the border with Kenya and sub-divided into 13 administrative units or words and one capital town Yabello. The total area of Borena zone is about 55,718 km². The zone is named after the majority ethnic group that has occupied the area and total population estimated 770,000. The agro-climate zone the area is subdivided into tropical, subtropical and desert. Borana zone is part of the Ethiopia's pastoral lowlands that occupy 60 per cent of the nation's landmass.¹¹ Rainfall reflects altitude – 900mm in the more mountainous north-west and 600mm in low land areas.



2.2 Data and methodology

In the study area OWWDSE has carried out intensive e ground exploration which includes geophysical investigation , imaging and test well drilling in the different locations of Borena zone ,during the period of 2006-2018) with budget allocated by the regional government and Pastoral area food security program development.

The objective of this study was to identify potential well site locations and to know the overall groundwater conditions of the study area using surface geophysical survey. More specifically, the study was aimed to understand the nature, number and type of aquifers within the study area so as to identify the potential well sites that could balance the water demand and supply. Accordingly there more than two hundred sites identified for study out these water quality and isotopes analysis was carried for human and livestock consumption with varying yield and water quality parameter standards.

Methodology

The data and methodology applied is as described steps below:

- Digital elevation model, DEM(SRTM 30m resolution
- Landsat 8 image OLI(February 2023)
- Ancillary data(geological, soil, climate(precipitation)data)
- Water quality laboratory analyzed data
- Software

Software	Uses	Source	Version	Remark
ArcGIS	Spatial data analysis(raster, vector, interpolation, mapping etc)	ESRI(crack)	10.7.1	
ENVI	Image analysis		5.3	
SNAP	InSAR data analysis	ESA(free)	9	
Golden surfer	Contour, Grid, Kriging interpolation		15	
SAGA GIS	Terrain, landforms		2.7	
QGIS			3.3.0	
PCI Geomatica				
Microsoft visio	Flow chart builder	Crack	2016	

Preprocessing

- Landsat 8(mosaicking, layer stacking, sub setting, radiometric and atmospheric correction)
- Water quality data(checking for data distribution(normality)
- Climate data(downscaling the CRU grid data to the local study area)
- Administrative, soil, land use land cover and geology shapefile clipped to the study area.

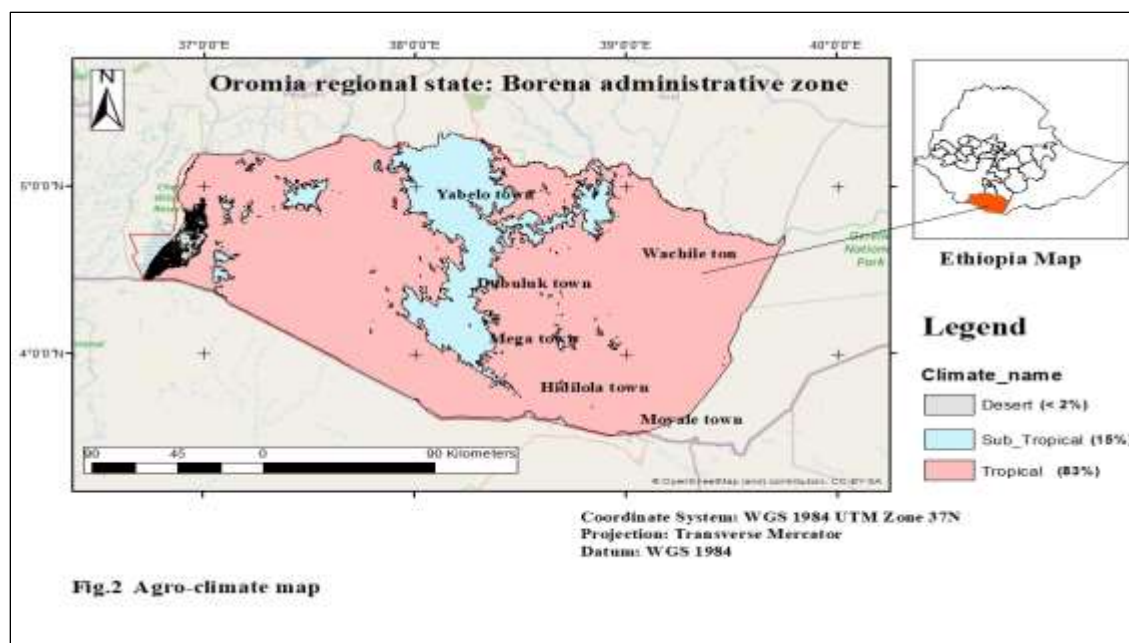
Processing

- Mapping of land use land cover, soil type, geology and climate zones
- Supervised classification, mapping and accuracy assessment.
- Geostatistical analysis and modeling

3. Borana agro climatic , ecology and pastoralism

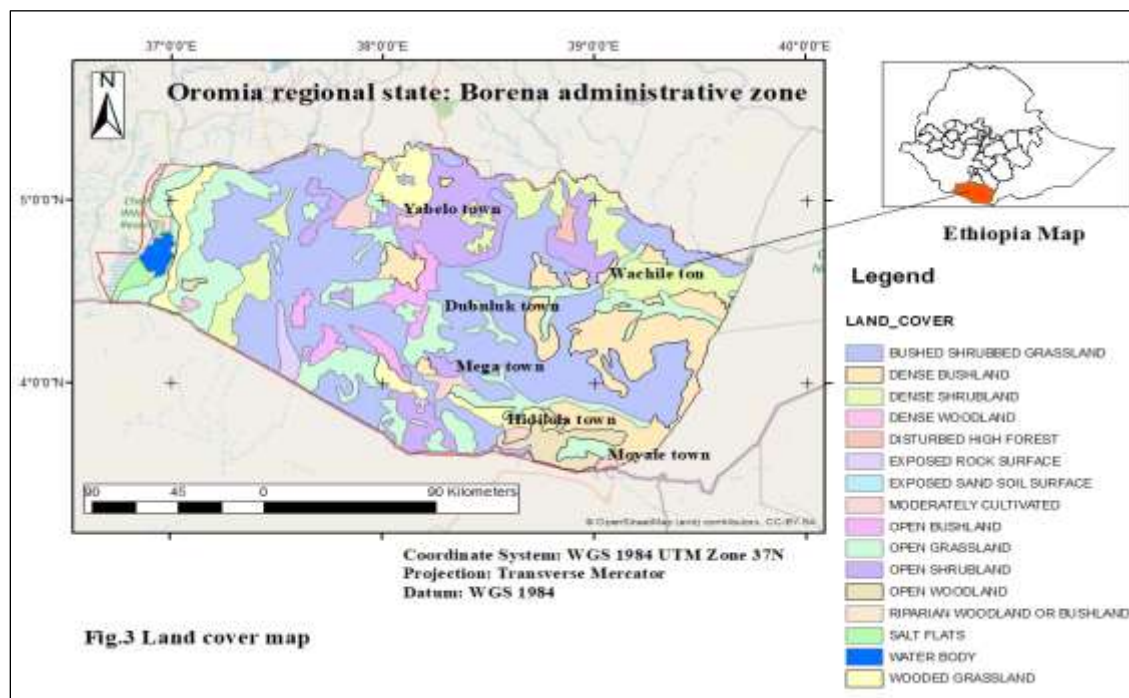
Borana zone is part of the Ethiopia's pastoral lowlands that occupy 60 per cent of the nation's landmass. Rainfall reflects altitude – 700mm in the more mountainous north-west and 500mm in lowland border area and is bi-modal: the spring ganna rains account for 60 per cent, while the shorter and more erratic autumn haggaya rains account for the balance. Rainfall is variable and erratic and drought not uncommon and this is reflected in the vegetation which is a rich mixture of open grasslands, perennial herbaceous and woody vegetation.

To utilize this spatially and temporally variable vegetation effectively requires the management of mixed herds – cattle and sheep for the grasslands and goats and camels for the woody species that are mobile and moved between wet and dry season grazing areas and drought reserves when the rain fails.



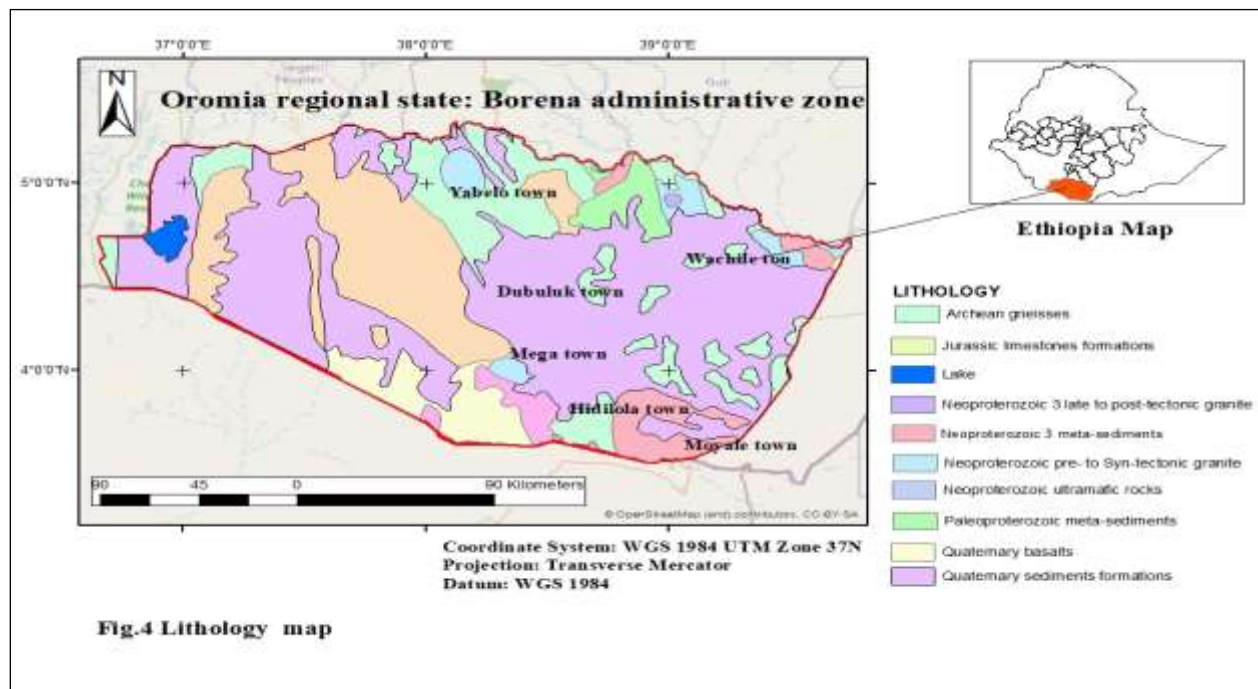
4. Land cover

The land cover of the area is mostly shrub grass land, shrubs and grass lands. As most of the area is occupied by pastoralist, there is no cultivated land but small Agricultural work is practiced near Yabello and Mega (Figure 3.5). According to Oromia water works design and study enterprise (OWWDSE) study, the western and north-west high lands of the project area and the low lands close to ridge escarpments are relatively well covered with mixed type of vegetation. The vegetation in these areas are dry lowland woods, mixed with hard wood coniferous Shrubs consisting of Podocarpus and Juniperous trees.



5. Geology

The geological setting of the vast tract of Borena land in areas south west of Yabello-Moyale main road and North of the Ethio-Kenyan border is found partly within the East African orogen, close to and partly in the broadly rifted zone of East African Rift system in the South Ethiopia. The geology of this area consists of two major groups of rocks named as Precambrian crystalline basement complex, Cenozoic volcanic and quaternary superficial deposits. The Precambrian crystalline Basement rocks which are exposed in the northeastern and southwestern parts of the area are comprised of high-grade gneisses, schist, weakly to moderately metamorphosed volcano-sedimentary rocks. The older metamorphic complex is formed from various gneiss and granulites invariably intruded by pre- and post-intrusive and younger plutonic rocks. The Precambrian rocks of the area exhibit similar lithologic association, structural features and grade of metamorphism characteristic to both Mozambique Belt and Arabian Nubian Shield to the south and north respectively (Aweke, H. and Hailu, F., 2007). The Cenozoic volcanic forms the mountain ridges of the western and low-lying plain areas in the central and southern parts whereas the eastern elevated grounds were formed from the basement rocks. The greater percentage of the area except the margins is characterized by subdued grounds of volcanic plains underlain by hundreds of meters thick basalt flows with fine to coarse columnar structure. Being bounded by the Ethiopia rift in northeast to southwest and its resulting volcanic mountains from west, the Borena land which is covered by the volcanic 5 plains is crossed by two rift systems called Ririba rift and Mega rift systems. The Ririba Rift has north-south trend whereas Mega rift system has northwest-southeast trend.



6. Hydrogeology

The difference in mineralogy, texture and structure of the volcanic rocks leads to have variable water bearing capacity. Groundwater circulation and storage in the volcanic rocks depend on the type of porosity and permeability formed during and after the rock formation (Alemayehu, T., 2006). The occurrence of groundwater in the basement and volcanic rocks is governed by the volume and behavior of faults, fissures, flow breccias, porous zones between successive lava beds, cracks and joints (Belete et al 2000). The primary porosities such as vesicles and flow contacts or interflow spaces in the volcanic rocks are also factors that determine the water bearing capacity in addition to the aforementioned secondary porosities. The groundwater occurrences in the

metamorphic rocks are in the fractured and weathered parts of the rock. In the case of volcanic rocks, the sizes of fractures are the predominant factors whereas for metamorphic rocks; the storage capacity is determined by the type of weathering products and the thickness of the consequent weathered layer. Metamorphic rocks are mostly impervious and serve as aquicludes and the only water bearing zones in many of them are the upper loose and weathered parts. Hence metamorphic rocks are characterized by secondary permeability which decreases with depth, the groundwater yield to wells decrease rapidly as depth increase.

7. Climate : rainfall and temperature

Rainfall play the most significant role in inducing flood (Pal and Samanta, 2011; Ali et al., 2019). Inverse Distance Weighted (IDW) algo- rithms (Othman et al., 2006; Chen and Liu, 2012) have been used in the ArcGIS environment to spatially map the rainfall. The rainfall map shows that the western portion of the research region experiences more rain than the eastern portion.

The rainfall data for this study was obtained from CRU TS (Climatic Research Unit gridded Time Series) which is a widely used climate dataset on a 0.5° latitude by 0.5° longitude grid over all land domains of the world except Antarctica. It is derived by the interpolation of monthly climate anomalies from extensive networks of weather station observations.

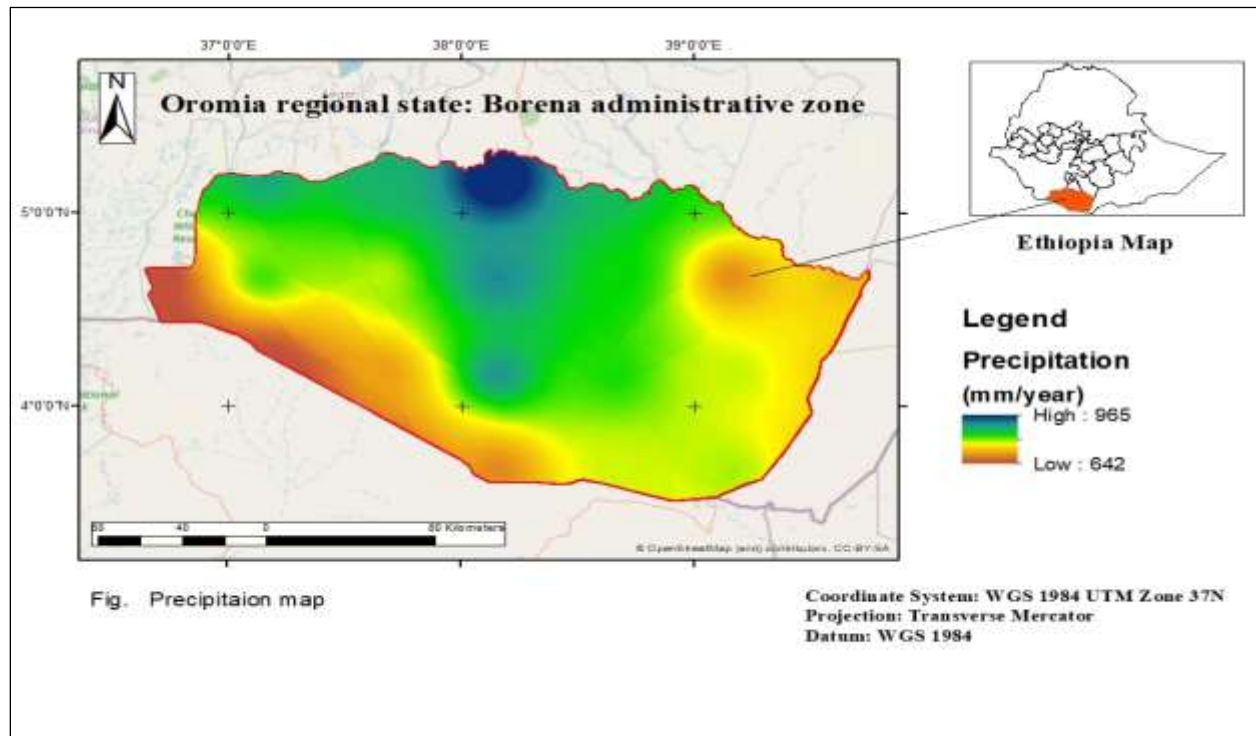


Fig. 5 Precipitation map of the study area

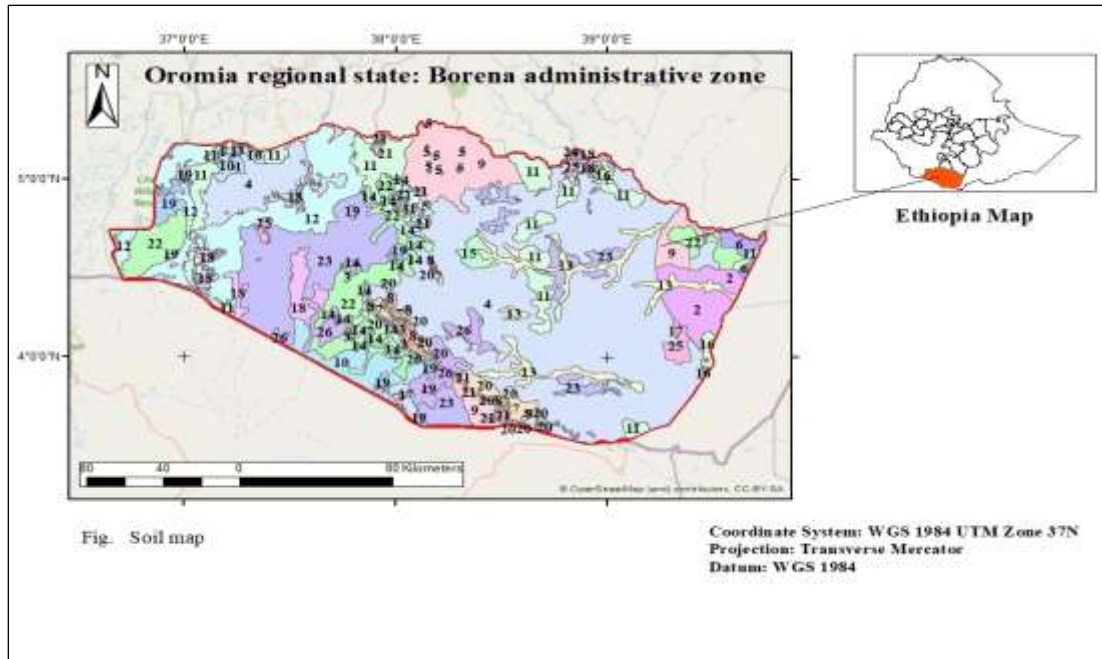


Fig.6 Geological map of the study area

Table 2: soil type and total area covered

ID	Soil type	Area_sq_km	Percentage
1	Calcic Cambisols	38.80	0.09
2	Calcic Xerosols	1,279.70	2.91
3	Cambisols	56.92	0.13
4	Chromic Cambisols	16,452.39	37.39
5	Chromic Luvisols	40.89	0.09
6	Chromic Vertisols	241.09	0.55
7	Dystic Nitosols	585.76	1.33
8	Dystric Cambisols	287.36	0.65
9	Euthic Nitosols	2,886.88	6.56
10	Euthric Regosols	1,210.98	2.75
11	Eutric Cambisols	3,178.55	7.22
12	Eutric Fluvisols	3,745.96	8.51

ID	Soil type	Area_sq_km	Percentage
13	Eutric Vertisols	1,357.55	3.09
14	Gypsic Yermosols	646.32	1.47
15	Haplic Phaeozems	232.96	0.53
16	Haplic Xerosols	242.62	0.55
17	Lakes	12.44	0.03
18	Leptosols	1,295.59	2.94
19	No Soil	764.10	1.74
20	Orthic Acrisol	177.60	0.40
21	Orthic Luvisols	216.88	0.49
22	Orthic Solonchaks	3,004.20	6.83
23	Pellic Vertisols	4,625.53	10.51
24	Phaeozems	2.00	0.00
25	Vertic Cambisols	345.18	0.78
26	Vitric Andosols	1,074.28	2.44

Cambisol, one of the 30 soil groups in the classification system of the Food and Agriculture Organization (FAO). Cambisols are characterized by the absence of a layer of accumulated clay, humus, soluble salts, or iron and aluminum oxides. They differ from unweathered parent material in their aggregate structure, colour, clay content, carbonate content, or other properties that give some evidence of soil-forming processes. Because of their favourable aggregate structure and high content of weatherable minerals, they usually can be exploited for agriculture subject to the limitations of terrain and climate. Cambisols are the second most extensive soil group on Earth, occupying 12 percent of the total continental land area—mainly in boreal polar regions, in landscapes with high rates of erosion, and in regions of parent material resistant to clay movement. They are not common in humid tropical climates.

In order for a soil to qualify as a Cambisol, the texture of the subsurface horizons must be sandy loam or finer, with at least 8 percent clay by mass and a thickness of 15 cm (6 inches) or more. These soils naturally form on medium- to fine-textured parent materials under any climatic, topographic, and vegetative-cover conditions. They differ from Leptosols and Regosols by their greater depth and finer texture and are often found in conjunction with Luvisols. (FAO)

8. Image classification (supervised, MLH classifier)

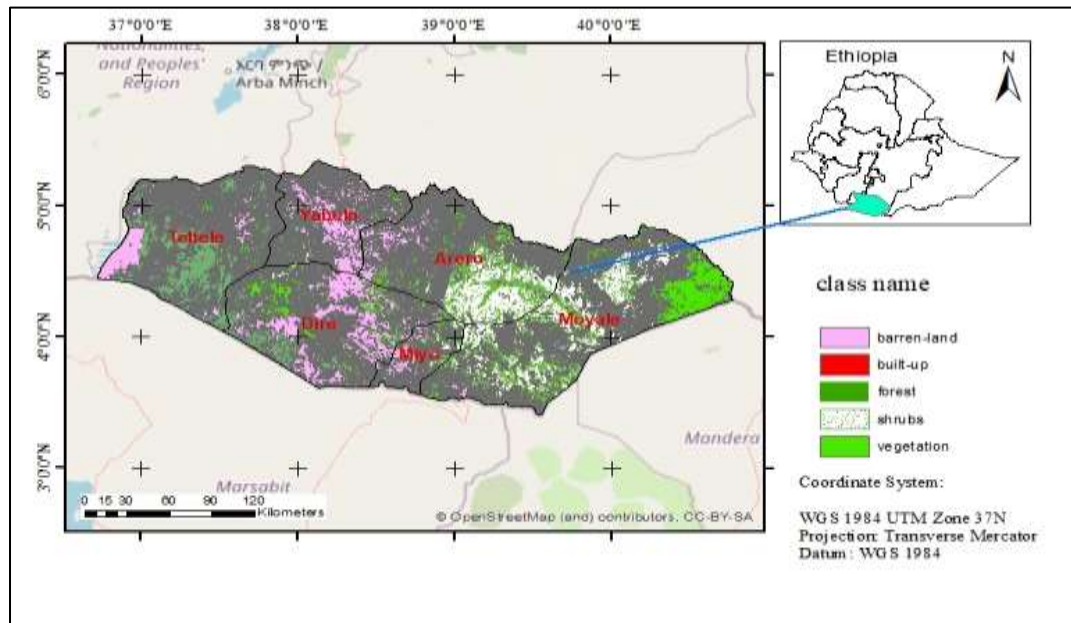


Fig. 7 Precipitation map of the study area

Table 3: classification table

ID	class name	area_sq_km					percentage (%)		
1	barren-land	16,464.13					29.55		
2	built-up	874.05					1.57		
3	forest	7,948.05					14.27		
4	shrubs	14,656.20					26.31		
6	vegetation	15,771.53					28.31		

ID	Class name	forest	vegetation	barren land	shrubs	built up	Total	User Accuracy	Kappa
1	forest	14	5	1	2	0	22	0.64	
2	vegetation	0	21	0	2	0	23	0.91	
3	barren land	0	5	27	0	0	32	0.84	
4	shrubs	0	4	0	28	0	32	0.88	
5	built up	0	0	0	1	3	4	0.75	
	Total	14	35	28	33	3	113		
	Producer Accuracy	1.00	0.60	0.96	0.85	1.00		0.82	
	Kappa								0.77

9. Groundwater supply, Water quality and Geostatistical analysis /modeling

Ground water sources

Groundwater is one of the drinking water supply source most part of Ethiopia, in Rural and Urban water supply system. The reason why ground water the first choice for drinking purpose it that most of surface supply sources such as spring water, surface water dams and rain water harvesting are subjected climate change can easily get deteriorated in quality and quantity even dry up in most cases due to hot weather and dry wind. Surface source is also susceptible to environmental pollution such as industrial and urban waste, agricultural fertilizer and pesticides chemical and the likes. Almost all the surface water sources must be treated chemically, physically and bacteriologically before supply for consumption.

Most of the ground water sources (sub-surface water) is potable for consumption without or with minimum treatment. The groundwater source is most of the time stable in quantity and uniform in temperature except in some cases where over extraction may cause; depletion, water table draw down, land subsidence and well abandonment. Moreover, groundwater source requires deep well drilling varying from shallow depth of 150m to deep depth of 400m. The investment cost for drilling and pumping the water from the deep depth (which includes electro-mechanical works, procurement and installation of submersible pumps with its accessories) is very high.

Groundwater quality

For any water supply system, the water quality; with its constituent of chemical, bacteriological and physical parameters must be tested in a laboratory and checked against the drinking water quality standards set by national health institutions or WHO standards. The major water quality parameter are; PH, Turbidity, Temperature, color, Total dissolved solids(TDS), Chlorides(Cl), Bicarbonates(HCO_3), sulphates(SO_4), Nitrates(NO_3), Magnesium(Mg), Calcium(Ca), Potassium(K), sodium(Na), Electrical conductivity(EC), Hardness(CaCO_3 , NaCO_3 etc.), Fluoride(F), Radioactive and Isotopes etc. Groundwater may get affected or contaminated due to groundwater movement(up and down), base flow, infiltration of surface water carrying washed chemicals from; acidic soil and limestones, polluted points sources, mineral ore rocks with heavy metals, fertilizer and pesticides.

Borena water supply sources

Borena zone is located in Oromia region, south Ethiopia along the Kenya boarder. Most of the settlers are pastoral communities of Borena (Ethnic Oromos). Their livelihood depend livestock (camels, cattle, and goats) rearing. The area is characterized by tropical climate and rainfall ranging from 650 to 950 mm per year. The area prone known by frequent drought, sometimes series which result in livestock mass death due to lack of water and pasture.

The water supply sources for human and animal water consumption is surface water ponds(Elas) and ground water supply.

Groundwater exploration

The Oromia regional government in collaboration with the Federal government has been since conducts ground water exploration project since 2011. The office in charge of this duty is Oromia water works design and supervision enterprise(OWWDSE).The objectives of the project is to undertake drilling of test well or production wells in selected groundwater potential sites with the help of geophysical investigation and surface survey tools. The goal of the project is to:

- Locate groundwater with sufficient quantity (yield).
- To conduct water quality testing to determine its potable or suitable for drinking (against water quality standards).
- Mapping the borehole distribution (points).

Table 4: schema of the data

Easting (m)	Northing (m)	Site name	Source Type	Elevation (m)	pH	EC (mS/cm)	T(°c)	Total hardness as CaCO ₃ (mg/l)	Total Alkalinity as CaCO ₃ (mg/l)
TDS (mg/l)	Na ⁺ (mg/l)	K ⁺ (mg/l)	Ca ⁺⁺ (mg/l)	Mg ⁺⁺ (mg/l)	Cl ⁻ (mg/l)	NO ₃ ⁻ (mg/l)	F ⁻ (mg/l)	HCO ₃ ⁻ (mg/l)	SO ₄ ⁻ (mg/l)

Geostatistical Analysis and Modeling(Hotspot water quality parameters analysis)

In this study two water quality parameters namely; **Electrical conductivity (EC) and Total dissolved solids(TDS)** were selected for Geostatistical analysis and modeling/mapping/ because of the following reasons;

- The two parameters water quality laboratory analysis result indicates is very high concentration (i.e. above the standards).
- The two water quality parameters are correlated and can usually expressed by a linear equation such as:
$$EC \text{ (cm/ } \mu\text{S)} \text{ (@ } 25^{\circ}\text{c)} = a + TDS(\text{mg/l}) * b$$
- These two parameters are indicators of salinity level which make them useful as one way of studying other sources of salinity intrusion.
- Conducting study and analysis of TDS and EC is very important because both can more illustrate Groundwater quality.

Bivariate Analysis:

Bivariate analysis is the understanding and quantifying of the relationship two variables. In our case TDS and EC are the two variables(parameters) selected for the reasons cited above.

Table 5: data statistics

Descriptive Statistics										
ID		N	Range	Minimum	Maximum	Sum	Mean	Std.	Variance	
		Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error		
1	pH	89	2.27	6.33	8.60	668.25	7.51	0.05	0.47	0.22
2	EC (mS/cm)	89	11,536.00	144.00	11,680.00	158,208.00	1,777.62	210.42	1,985.10	3,940,609.10
3	T(oc)	89	41.40	-	41.40	2,215.10	24.89	0.81	7.65	58.51
4	Total hardness as CaCO ₃ (mg/l)	89	2,869.00	11.00	2,880.00	43,810.50	492.25	53.11	501.00	251,005.43
	Total Alkalinity as CaCO ₃ (mg/l)	88	3,036.00	24.00	3,060.00	31,620.96	359.33	48.18	451.99	204,297.25
	TDS (mg/l)	89	8,746.00	80.00	8,826.00	108,688.00	1,221.21	160.47	1,513.87	2,291,792.85
7	Na+ (mg/l)	89	1,888.50	11.50	1,900.00	17,583.00	197.56	32.00	301.93	91,162.71
8	K+ (mg/l)	89	373.70	1.30	375.00	2,234.60	25.11	5.44	51.34	2,635.77
9	Ca++ (mg/l)	89	701.30	2.70	704.00	9,574.38	107.58	10.68	100.74	10,148.41
10	Mg++ (mg/l)	89	733.30	1.10	734.40	5,522.69	62.05	10.96	103.38	10,686.52
11	Cl- (mg/l)	89	2,321.40	8.60	2,330.00	19,544.41	219.60	40.01	377.43	142,454.53
12	NO ₃ -- (mg/l)	89	44.50	0.10	44.60	576.68	6.48	1.00	9.42	88.65
13	F- (mg/l)	89	7.80	-	7.80	115.27	1.30	0.13	1.20	1.44
14	HCO ₃ - (mg/l)	89	3,703.90	29.30	3,733.20	41,456.31	465.80	65.49	617.85	381,735.53
15	SO ₄ -- (mg/l)	89	1,691.50	0.50	1,692.00	22,057.22	247.83	36.53	344.59	118,740.55

Geostatistical Analysis: EC and TDS

Spatial Continuity

Spatial Continuity is the correlation between values over distance.

No spatial continuity – no correlation between values over distance, random values at each location in space regardless of separation distance.

Homogenous phenomenon have perfect spatial continuity, since all values are the same (or very similar) they are correlated.

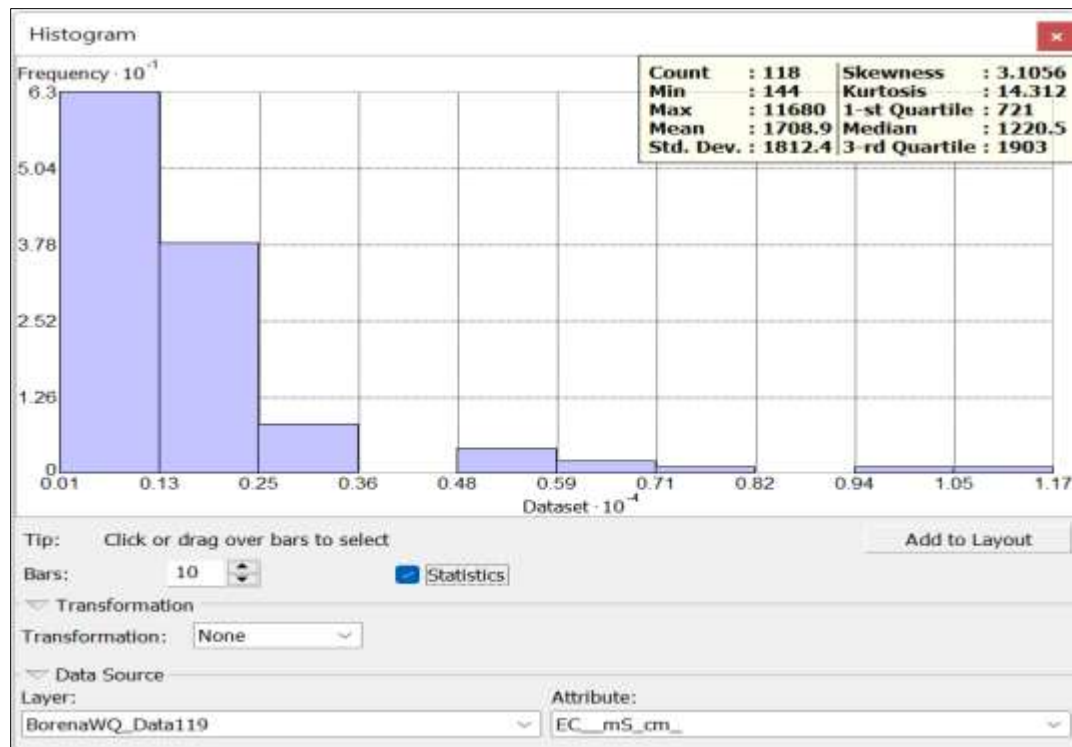


Fig.8 Electrical conductivity original data(positivity skewed distribution)

Checking Data Distribution

Checking data distribution is a preliminary step in the application of geostatistics such as Kriging in which it is assumed that the data are normally distributed over the study area. Distribution checking is done to guarantee this property for the data. In case the distribution is not normal, a data transformation method is applied to make the distribution normal. An ideal normal distribution should have 3 properties:

- 1) Mean and Median values of data should be as the same as possible,**
- 2) Data Skewness should be close to 0 and**
- 3) Data Kurtosis should be close to 3.**

In this study, the distributions are controlled using the Histogram tool and it's found that none of the parameters has a normal distribution, therefore, suitable Transformation methods were applied to the data. The results of the data checking process have been shown in Fig. below.

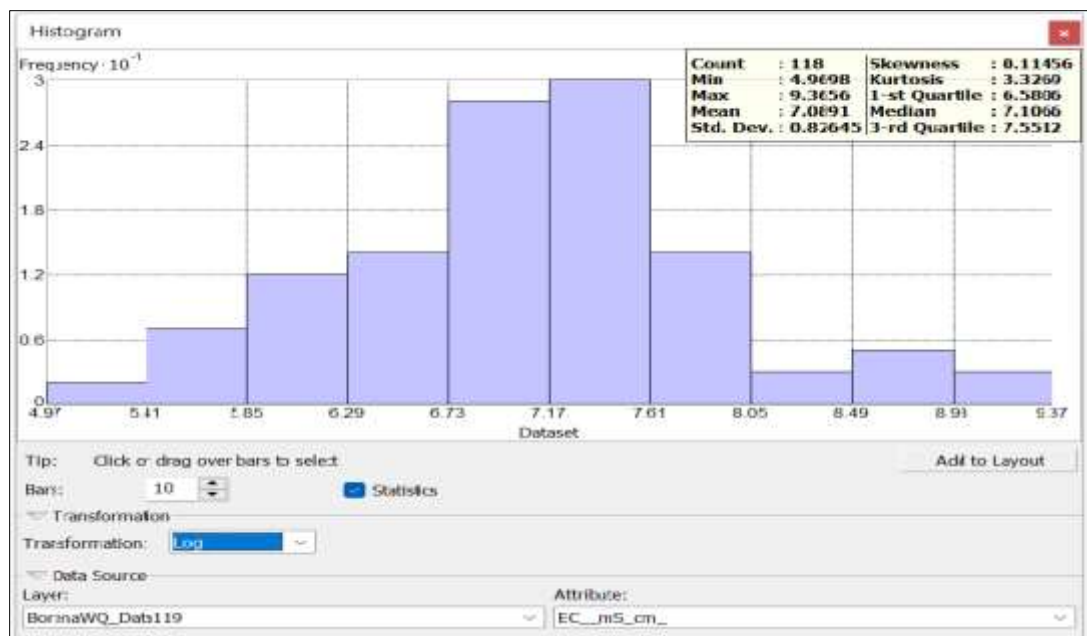


Fig. 9 EC transformed normal distribution

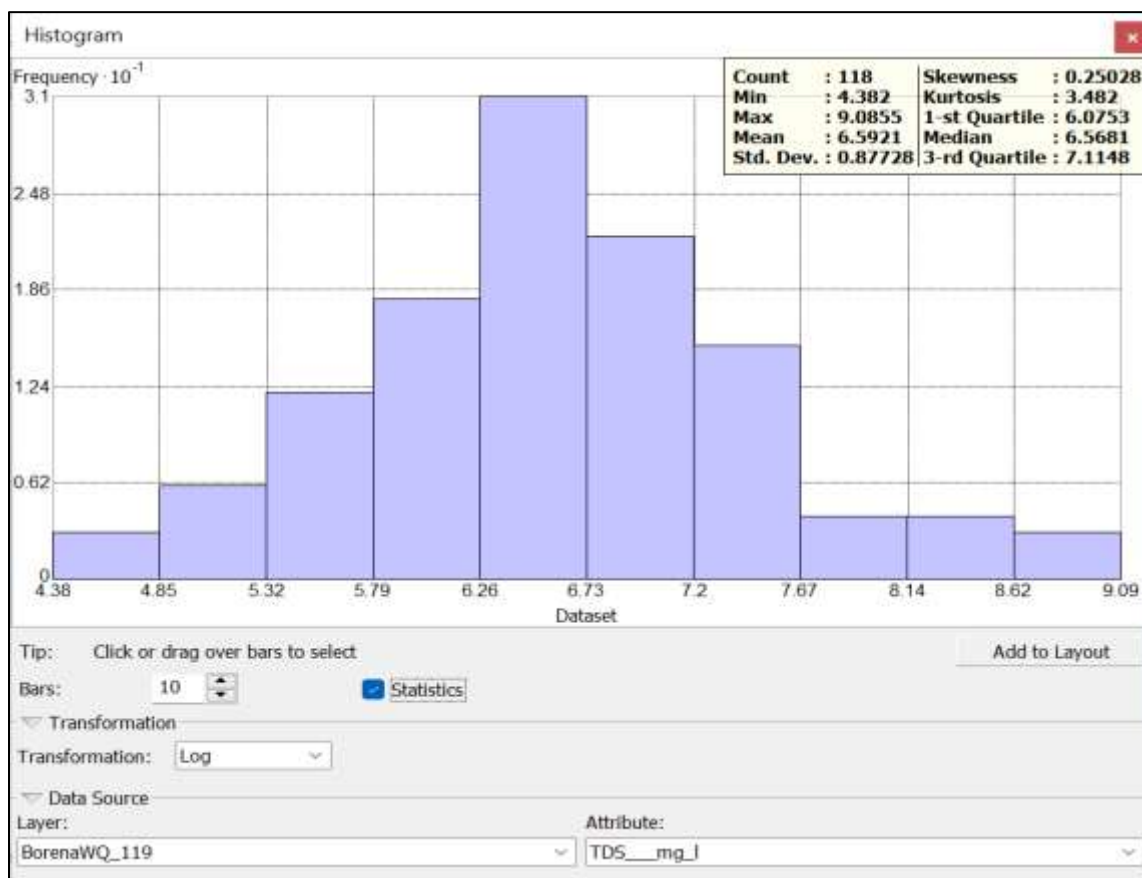


Fig.10 TDS after log transformed : Normal distribution with mean & median similar

QQ-plot of Electrical conductivity(EC)

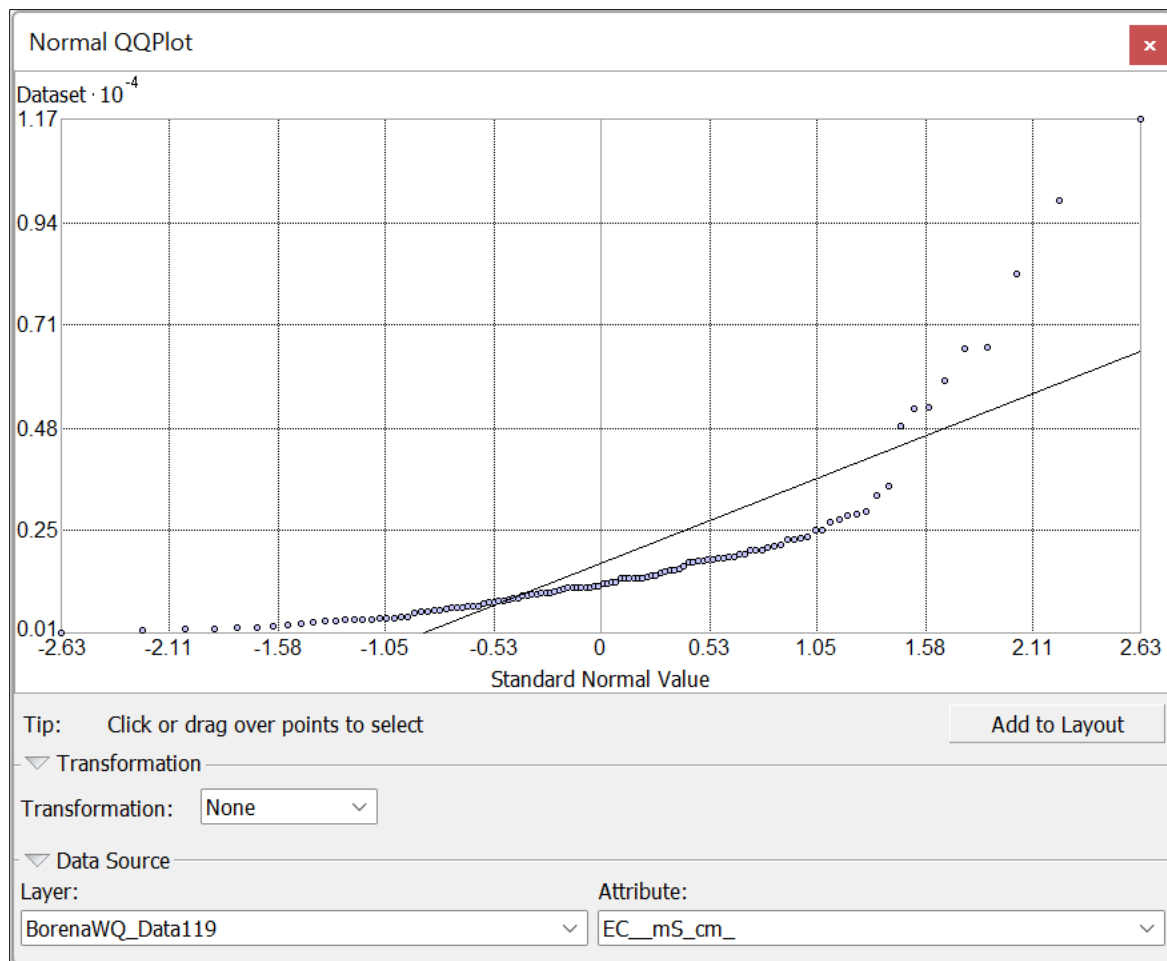


Fig. 11 QQ-plot (original data)

From the normal QQ plot above, it can be seen that the plot is not very close to being a straight line. The main departure from this line occurs at middle values of EC concentration. For the data that does not exhibit a normal distribution in either the histogram or normal QQ plot, it is necessary to transform the data to make it conform to a normal distribution before using certain kriging interpolation techniques. Accordingly, both parameters were transformed to get a normal distribution.

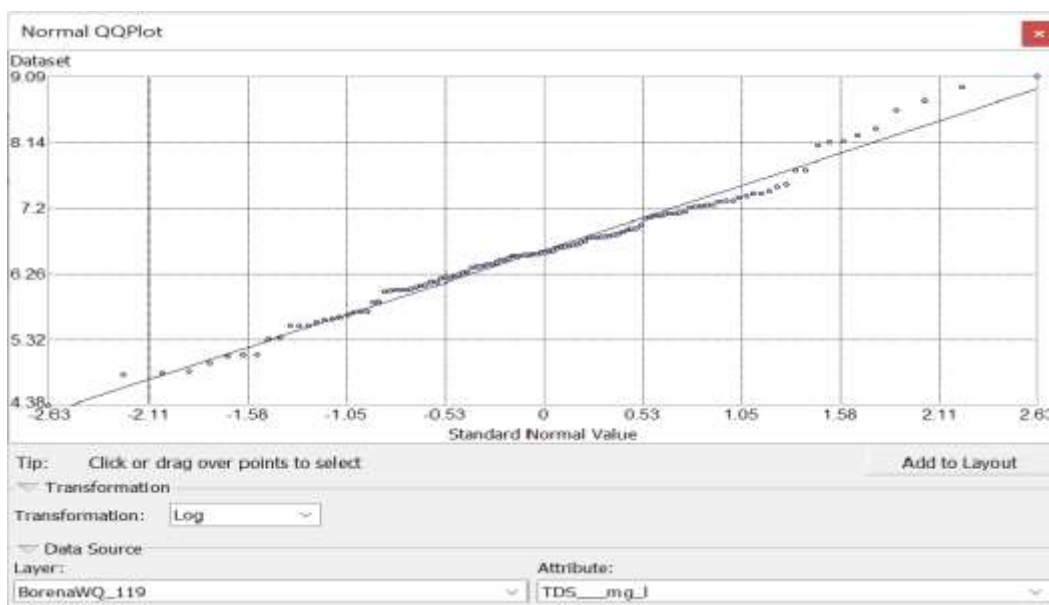
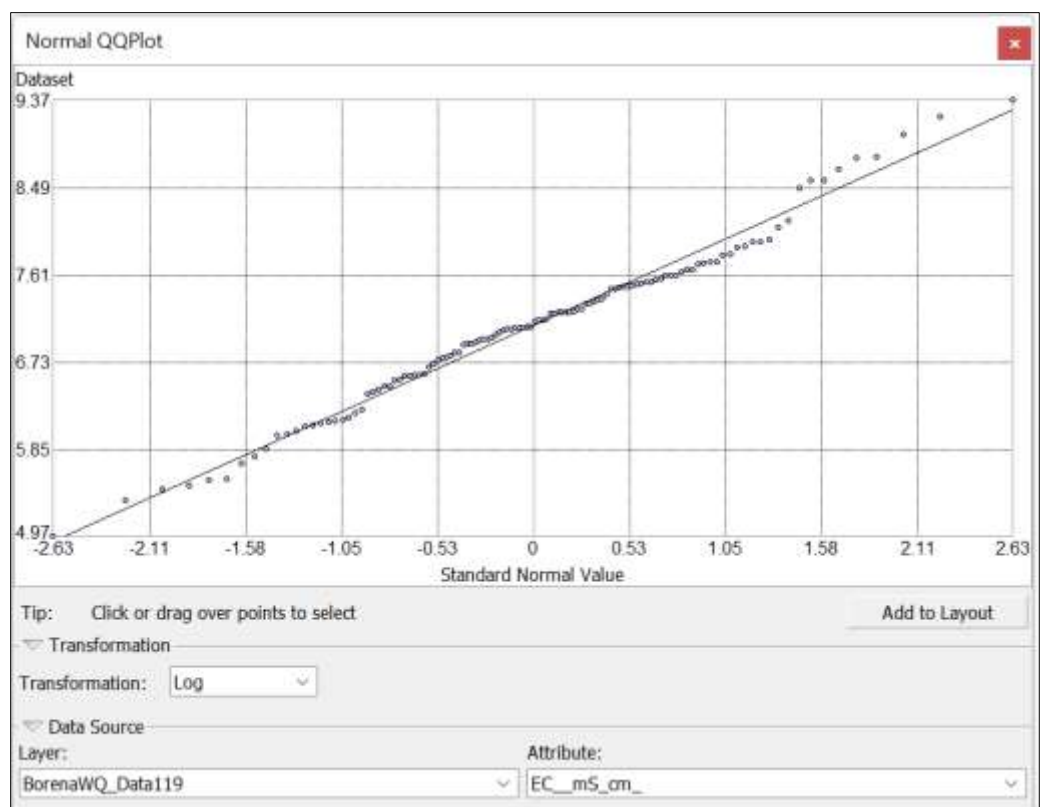


Fig. 12 TDS and EC Q-Q plot After log transformation

Semivariogram

The semivariogram/covariance cloud allows you to examine the spatial autocorrelation between the measured sample points. It is generally assumed that things that are close to one another are more alike.

The semivariogram/covariance cloud lets you examine this relationship. To do so, a semivariogram value, which is the difference squared between the values of each pair of locations, is plotted on the y-axis relative to the distance separating each pair of measurements, which is plotted on the x-axis. Each red dot in the semivariogram/covariance cloud represents a pair of locations. Since locations that are close to each other should be more alike.

In the semivariogram plot, the locations that are closest (on the x-axis) should have small semivariogram values (low values on the y-axis). As the distance between the pairs of locations increases (moving up from the x-axis), the semivariogram values should also increase (move up on the y-axis) as shown below.

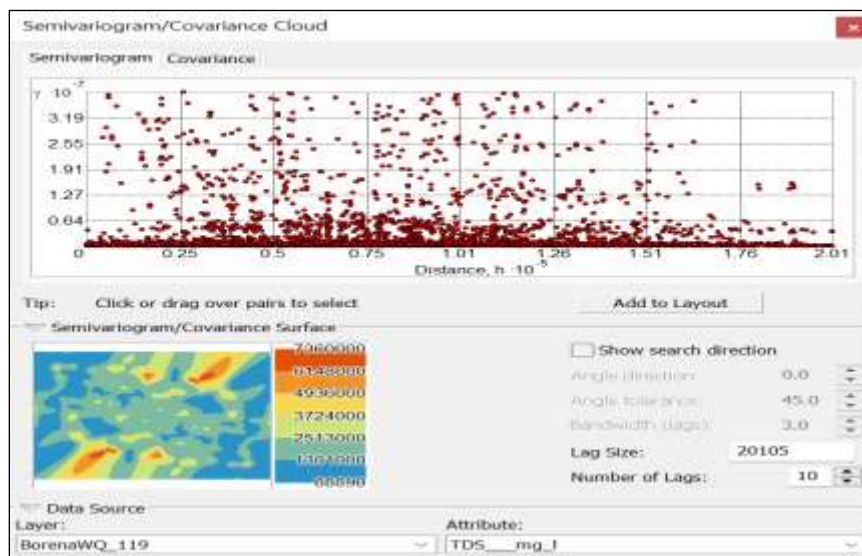
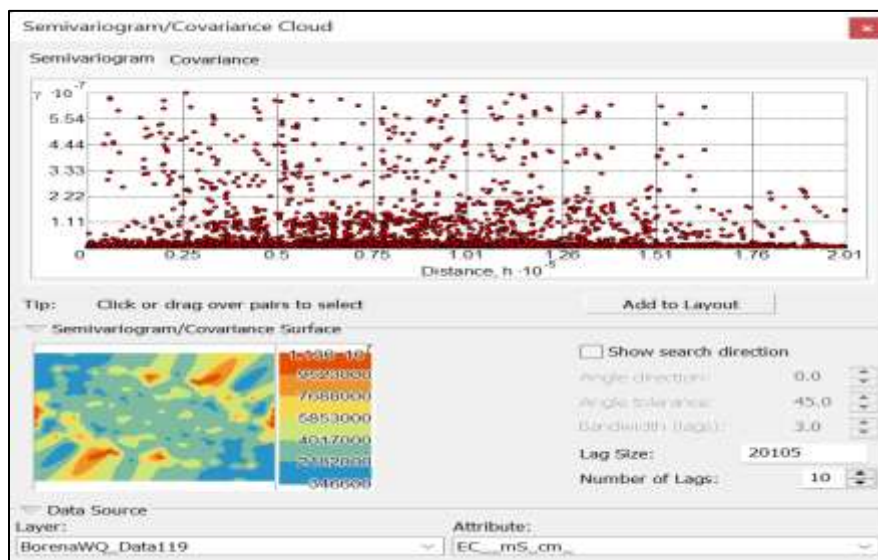


Fig. 13 TDS and EC semi-variogram diagram

Covariance diagram

Covariance measures the direction of the relationship between two variables. A positive covariance means that both variables tend to be high or low at the same time. A negative covariance means that when one variable is high, the other tends to be low. In this analysis the two parameters demonstrated similar directional similarity.

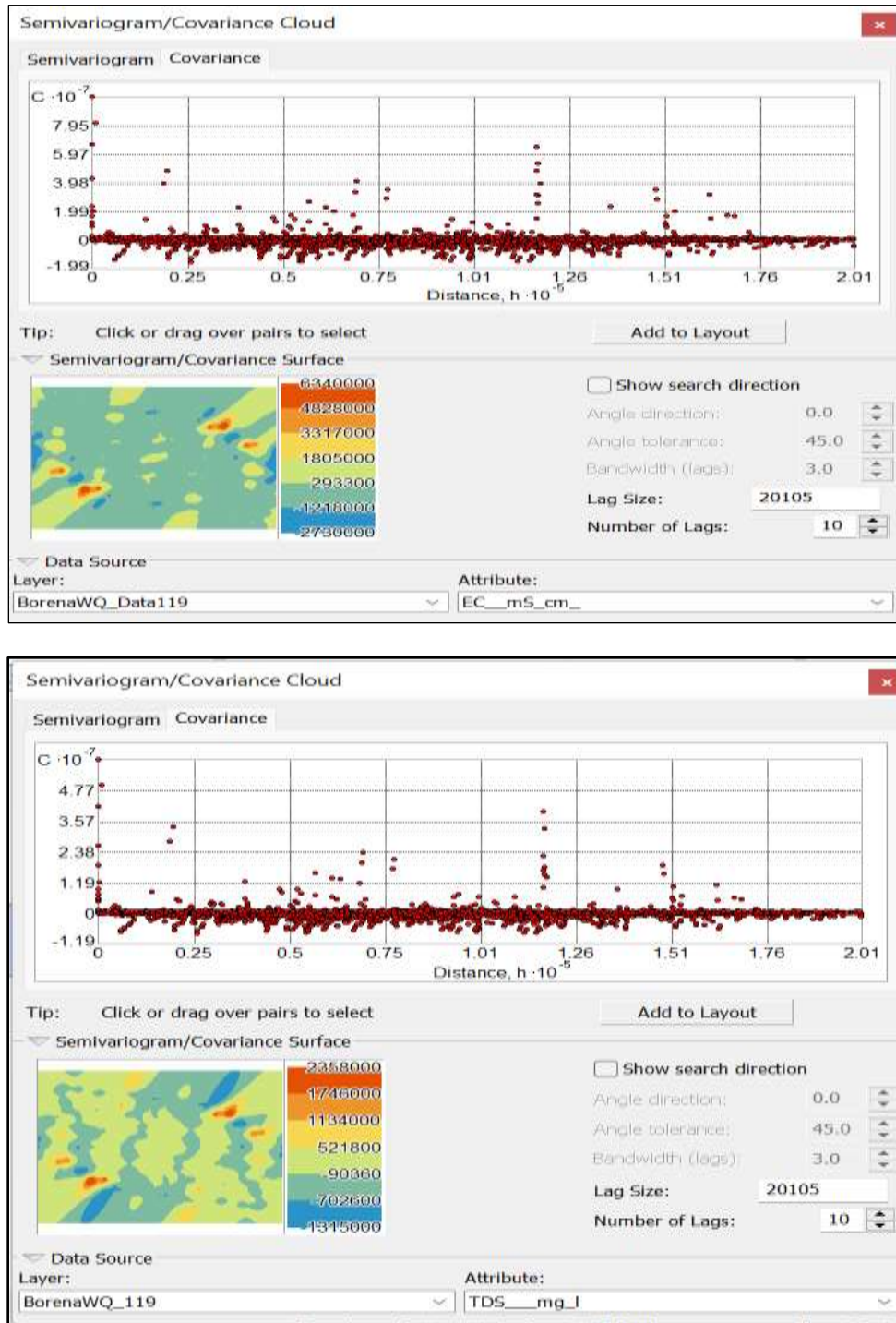


Fig. 14 TDS and EC covariance After log transformation

Ordinary Kriging Interpolation Method

As a powerful statistical interpolation method used for diverse applications such as health sciences, geochemistry and pollution modeling, Kriging assumes that the distance or direction between sample points reflects a spatial correlation that can be used to explain variation in the surface. Kriging technique is an exact interpolation estimator used to find the best linear unbiased estimate.

Ordinary Kriging is a spatial estimation method where the error variance is minimized. Ordinary kriging assumes the model: $Z(s) = \mu + \varepsilon(s)$;

Where μ is an unknown constant. To use this interpolation model, data must be normally distributed as well as being auto-correlated.

In the present study after exploring the dataset of each qualitative parameter, an appropriate method of Ordinary Kriging was used to create map layers of the parameters. Before using the interpolation techniques, it's necessary to explore data to check their spatial dependence and distribution. In the Geostatistical Analyst, Exploratory Spatial Data Analysis graphs (ESDA) are used for examining data prior to interpolation. The Histogram and Semivariogram tools were used to check the distribution and autocorrelation of the data respectively.

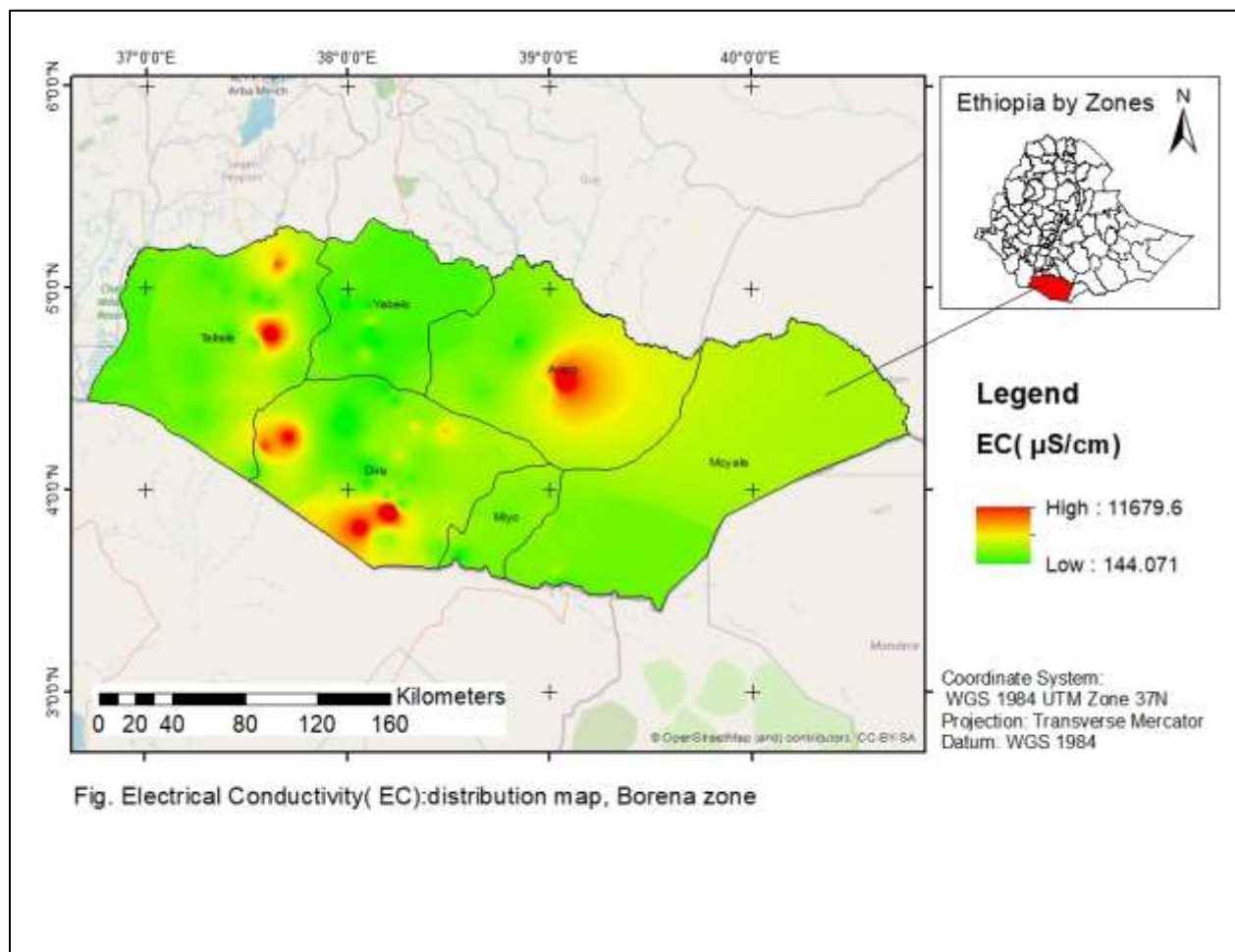


Fig.14.1

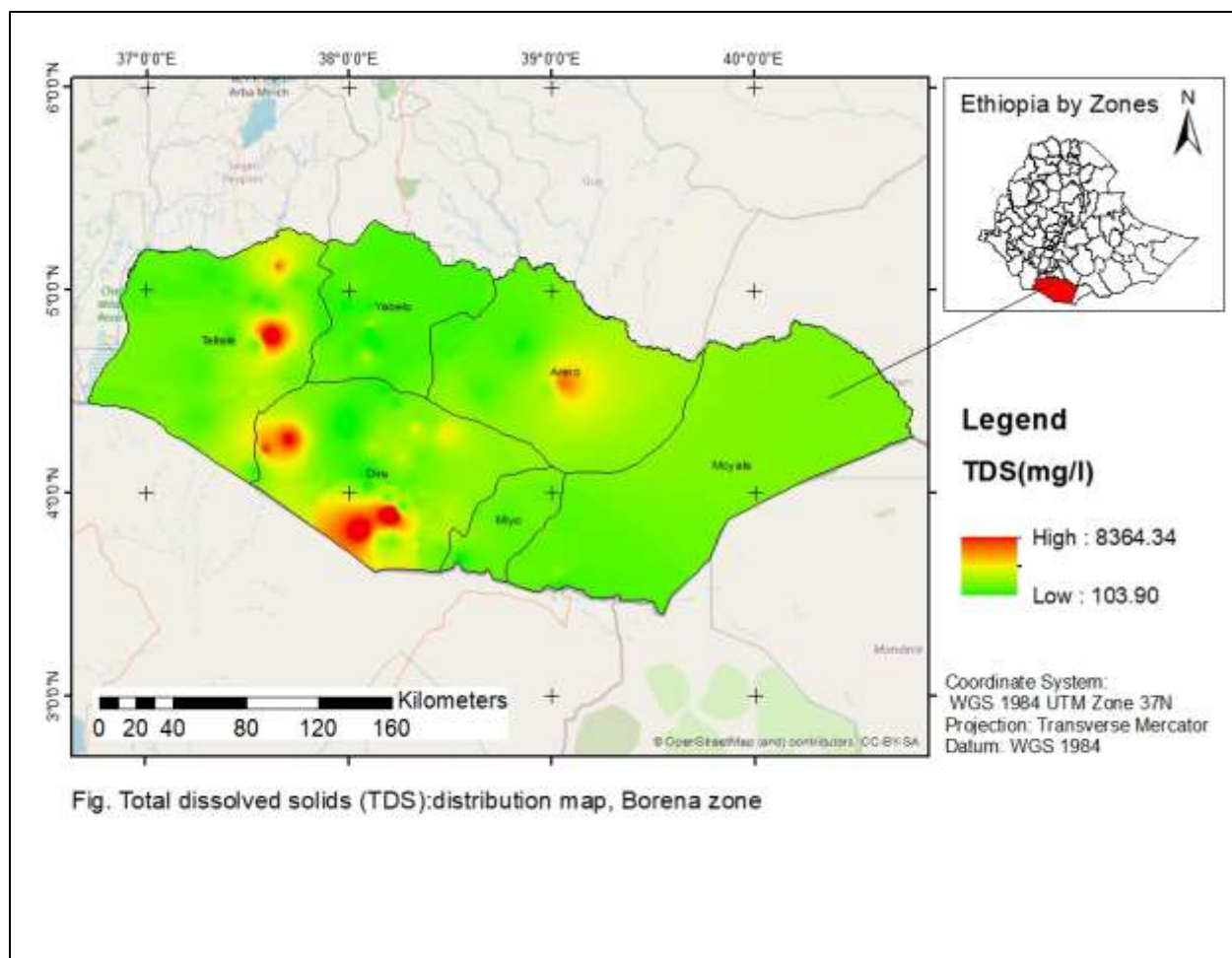


Fig.14.2

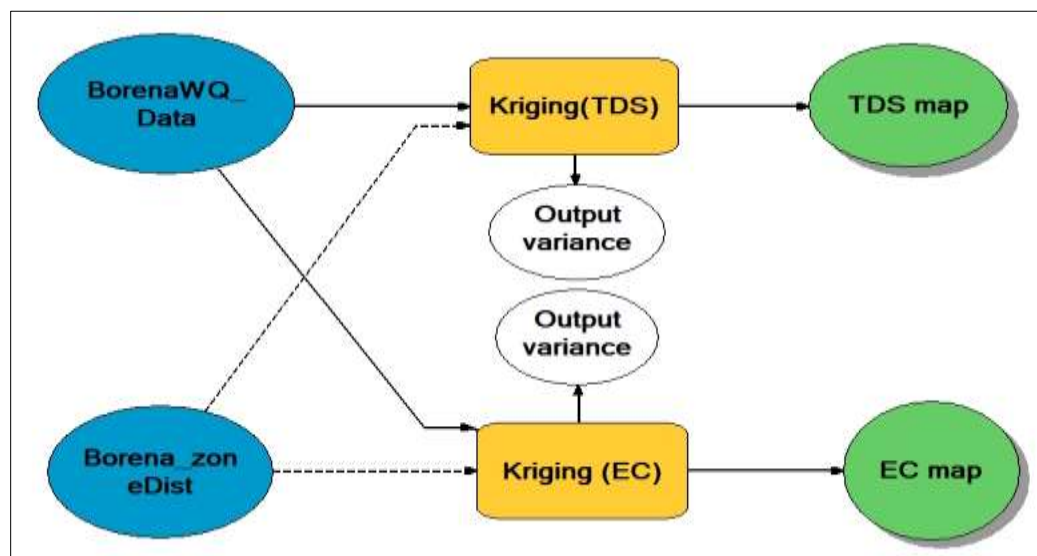


Fig.15 Kriging model builder

10. Application of Statistical Package for the Social Sciences(SPSS),

Is one of the most-used statistical analysis tools.

There are a variety of methods for each Kriging model to use in the Geostatistical Analyst but the best one is the method in which, the standardized mean error is close to 0, the root-mean square error and average standard error are as small as possible (this is useful when comparing models), and the root-mean-square standardized error is close to 1.

Descriptive Statistics			
	Mean	Std. Deviation	N
EC (mS/cm)	1708.907	1812.3603	118
TDS(mg/l)	1107.314	1356.5211	118

Model Summary ^b									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	Change Statistics			Sig. F Change
						F Change	df1	df2	
1	.988 ^a	.977	.977	275.6591	.977	4941.433	1	116	.000

a. Predictors: (Constant), TDS(mg/l)

b. Dependent Variable: EC (mS/cm)

Correlations			
		EC (mS/cm)	TDS(mg/l)
EC (mS/cm)	Pearson Correlation	1	.988**
	Sig. (2-tailed)		.000
	Sum of Squares and Cross-products	384304038.0	284327161.4
	Covariance	3284649.897	2430146.679
	N	118	118
TDS(mg/l)	Pearson Correlation	.988**	1
	Sig. (2-tailed)	.000	
	Sum of Squares and Cross-products	284327161.4	215297495.4
	Covariance	2430146.679	1840149.533
	N	118	118

** . Correlation is significant at the 0.01 level (2-tailed).

** . Correlation is significant at the 0.01 level (2-tailed).

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	TDS(mg/l) ^b	.	Enter

a. Dependent Variable: EC (mS/cm)

b. All requested variables entered.

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	375489434.2	1	375489434.2	4941.433	.000 ^b
	Residual	8814603.747	116	75987.963		
	Total	384304038.0	117			

a. Dependent Variable: EC (mS/cm)

b. Predictors: (Constant), TDS(mg/l)

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error				Lower Bound	Upper Bound
1	(Constant)	246.561	32.813		7.514	.000	181.570	311.553
	TDS(mg/l)	1.321	.019	.988	70.295	.000	1.283	1.358

a. Dependent Variable: EC (mS/cm)

EC=2.46.561 + 1.321 * TDS..... is the linear equation relating the two parameters

Coefficient Correlations^a

Model		TDS(mg/l)
1	Correlations	TDS(mg/l)
	Covariances	TDS(mg/l)

a. Dependent Variable: EC (mS/cm)

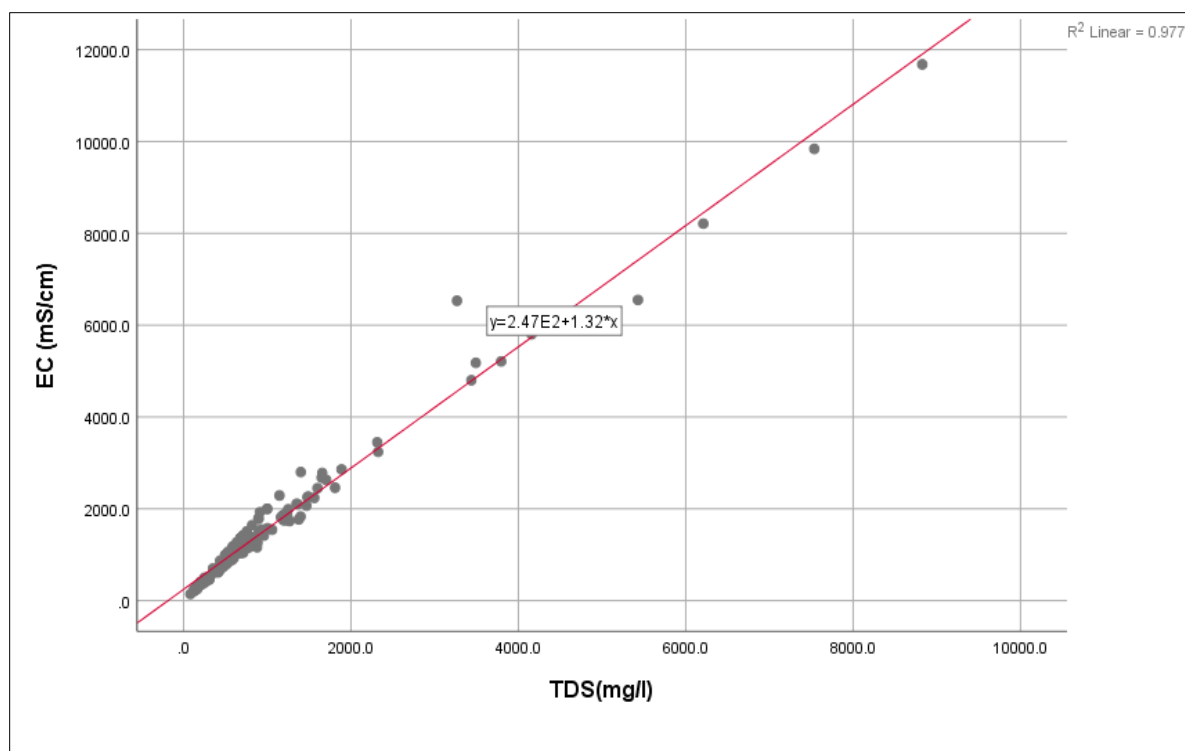


Fig. 16 scatter graph(dot)

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	352.211	11902.394	1708.907	1791.4551	118
Residual	-866.2321	1972.2789	.0000	274.4786	118
Std. Predicted Value	-.757	5.690	.000	1.000	118
Std. Residual	-3.142	7.155	.000	.996	118

a. Dependent Variable: EC (mS/cm)

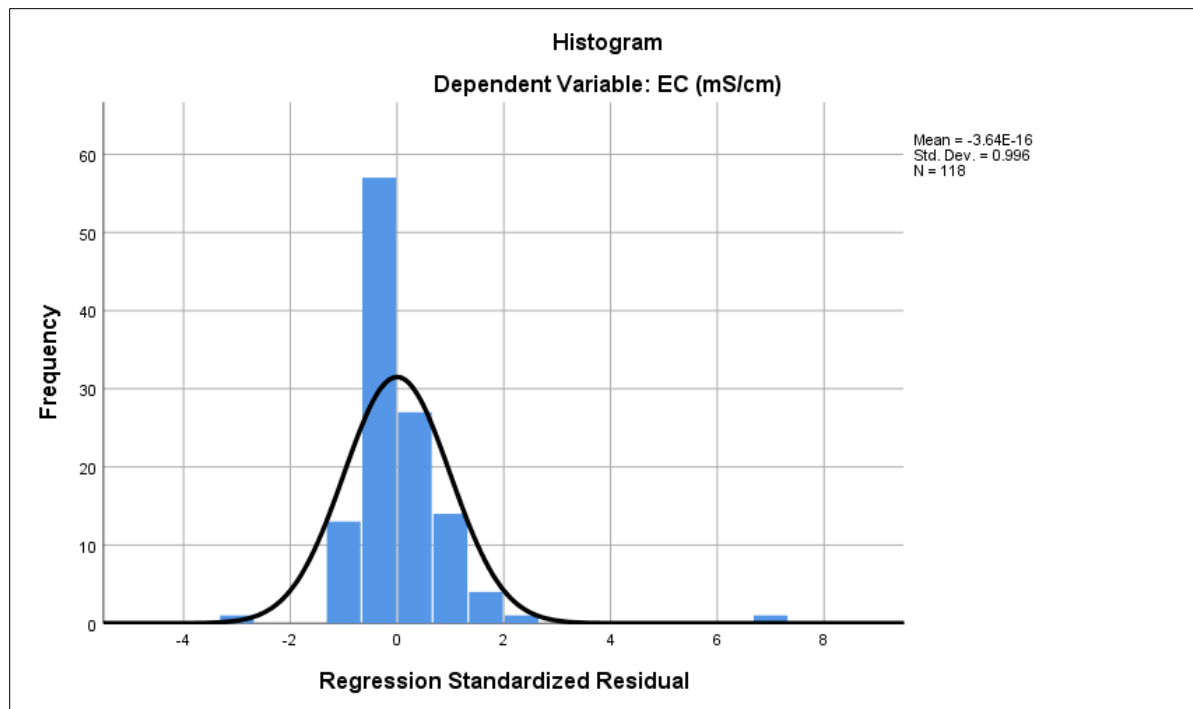


Fig. 17

P-P Plot

P-P plot (probability–probability plot or percent–percent plot or P value plot) is a probability plot for assessing how closely two data sets agree, or for assessing how closely a dataset fits a particular model. It works by plotting the two cumulative distribution functions against each other; if they are similar, the data will appear to be nearly a straight line. This behavior is similar to that of the more widely used Q–Q plot, with which it is often confused.

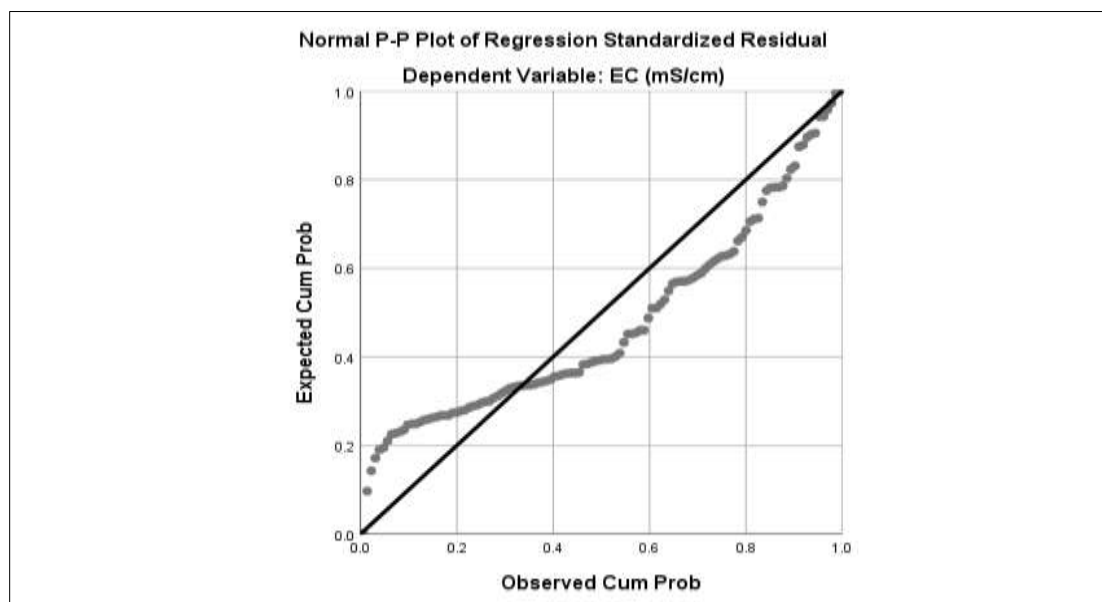


Fig.18

3. Challenges/ limitations

The following are challenges faced in this study;

- Due to wide study area(about 55,000 km²) satellite image mosaicing, layer stacking, band composite and image classification works is tedious and time consuming.
- Coarser spatial resolution of optical imagery made it difficult to discriminate feature cases such as rivers, roads, settlement and built-up areas during land use land cover image classification.
- Water quality parameter data lacked for wells.
- Field survey and validation based on google earth imagery

Discussions

TDS is the total amount of solids, in milligrams per liter that remain when a water sample is evaporated to dryness. It describes all solids (usually mineral salts) that are dissolved in water. The more salts are dissolved in the water; the higher is the value of ion concentrations and thus the higher electric conductivity.

Therefore, EC and TDS are related to one another. The EC against TDS from all sources of water measured in laboratory shows a straight correlation line ($R^2 = 0.997$) indicates that TDS accounts for some 97.7% of the variance in EC. That is, TDS predicts EC very well in this sample. Moreover, Electrical conductivity (EC) can be predicted in the scatterplot as **y** (shown on the y-axis) and **x is Total dissolved solids(TDS)** (on the x-axis). The linear relationship can be presented as;

$EC = 246.56 + 1.32 (TDS)$ best fit line equation

In the hotspot areas of Teltelle, Dire and Arero districts the concentration of total dissolved solid(TDS) and electrical conductivity (EC) much higher than the maximum allowable concentration (i.e. 1000 mg/L and 1500 μ S/cm) respectively (Ref. fig.14.1 and 14.2).

Fresh water is water that contains less than 1000 μ S/cm of electrical conductivity and less than 500 mg/l of dissolved solids is desirable for drinking and many industrial uses (WHO 2011). Water from deep wells from Yabello, Moyal and Miyo districts are classified low concentration in salinity i.e. low total dissolved solids and electrical conductivity concentration is potable water.

Conclusions

The study has presented an Geostatistical analyst tools(GAT) for modelling the relation between the groundwater quality parameters Total dissolved solids (TDS) and electrical conductivity(EC). It has been observed that the GAT best performed in analysis and predicting the spatial relationship and modelling electrical conductivity based on the total solids dissolved in the groundwater. It was revealed from the spatial and statistical analysis result of Electrical Conductivity (EC) and Total Dissolved Solids, the two samples of groundwater quality parameters are highly correlated and their relationship is linear; **$EC = 246.56 + 1.32 (TDS)$.**

Therefore, the calculation of TDS value based on EC value is very useful in doing water quality research. Numerous research have been conducted to find out the relationship between these two variables. This situation highly depends on water salinity and material contents. Some studies result argues and present that the relationships between the two parameters is ratio based and the relationship between TDS and EC is not always linear.. The higher the salinity level or material contents, the more complex mathematical equations needed in describing those parameters. The strongest correlation between TDS and EC is found in natural water. Lastly, the acquisition of TDS from EC conversion can be conducted in explaining general condition of water quality. However, for deeper analysis, the TDS concentration is better conducted in a laboratory by applying gravimetric analysis and the influence of salinity and other parameters must be studied.

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